

Breaking Bellman:

Dark Uncertainty and the Computable Cost of Desynchronized Decision Networks

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Abstract

No decision system operates alone. In practice, every model is a node in a network of decisions—one system’s output becomes another’s input. The Bellman equation is the foundation of sequential decision-making, and within any single system it may be perfectly specified. But at the boundaries between systems, where data crosses and codebooks do not, a form of uncertainty arises that no within-system diagnostic can detect. We call this *dark uncertainty*: structurally present, computationally consequential, and invisible from the inside.

Dark uncertainty is to decision networks what dark matter is to galaxies. It shapes outcomes. It may constitute the majority of real uncertainty in a networked system. And no instrument that operates within the system’s state space can see it—because the variable is not there.

We formalize dark uncertainty through a structural inversion of Kuramoto’s synchronization result. Strogatz (2003) and Kuramoto (1984) proved that coupled oscillators synchronize spontaneously because the coupling variable—phase—is in the signal. We prove the structural inverse: at serialization boundaries, the coupling variable—the codebook mapping symbols to meaning—is *not* in the signal. From this we derive that (i) for a fixed codebook-blind channel, all channel observables are invariant to codebook divergence, (ii) no coupling force acts on the codebook, and (iii) codebook divergence does not converge under data exchange alone, regardless of bandwidth, duration, or volume.

Any reduction in codebook ambiguity requires, in its causal history, at least one cross-boundary acquisition of information about the codebook—what we call *rotation*. To operationalize this at scale, we introduce a minimal, exhaustive decision interface for boundary coherence: **Halt** (boundary outside reasoning capacity), **Ask** (acquire cross-boundary information), **Act** (codebooks aligned, proceed). This interface composes across networked systems. We prove that boundary uncertainty behaves as a conserved cost: it can be reduced by acquiring information (Ask) or expressed as expected loss (Act without prior rotation), but not eliminated without payment. A decision modeler can get their model right. But if the network of decisions surrounding that model is desynchronized—if the codebooks have drifted and no one has measured the drift—the Bellman equation converges confidently to a number that does not correspond to reality. Dark uncertainty is the name for that structural gap. This paper formalizes it. All claims carry falsification criteria.

1 The Problem

No decision system operates alone.

Every reinforcement learning agent, every dynamic programming solution, every planning algorithm that optimizes over multiple steps exists within a network of other systems. It receives input from upstream systems. It sends output to downstream systems. Each handoff is a serialization boundary—a point where one system’s internal state is written into structured data for another to consume.

Within any single system, the Bellman equation may be perfectly specified. The state space is correct. The transition model is calibrated. The reward function captures what matters. The value function converges. A decision modeler can get their model right.

But the model is not the system. The system is the network. And at the boundaries of that network, something happens that no within-system diagnostic detects: the codebook—the mapping from symbols to real-world meaning—does not cross with the data. The downstream system receives syntactically perfect input, passes every validation check, and optimizes confidently over a state space that may be desynchronized from reality.

We call this *dark uncertainty*. Like dark matter, it is structurally present, shapes outcomes, and is invisible to every instrument that looks only at what is inside the system. You cannot detect it by examining the data. You cannot detect it by running diagnostics. You can only detect it by looking outward—across the boundary—and asking whether the system on the other side means the same thing you do.

Kuramoto’s coupled oscillators synchronize for free. Put thirty metronomes on a floating platform and within minutes they lock into perfect unison. The phase *is* the signal. Listening is sufficient. Above a critical coupling strength K_c , order emerges for free.

The question this paper answers: can codebook alignment happen the same way? Can the meaning synchronize passively through the shared medium of data exchange?

No. And the proof is short.

But first: this is not a theoretical problem.

2 The Problem in the Wild

In 2020, Colorado passed landmark legislation requiring every law enforcement agency in the state to report use-of-force data to a centralized system. The channel was built. Data flowed. Five years later, the data is largely useless [9].

Denver—the state’s largest department—reported three different numbers for the same year: 132 use-of-force contacts according to the state database (which only captured December), 3,000 according to a footnote in the same state report, and 844 according to the city’s own dashboard. Colorado Springs reported 101 to the state; their own records show 668. Monument reported zero to the state; their internal records show two.

The channel works. Every record arrives. The data validates. And none of it means the same thing.

The state’s Division of Criminal Justice acknowledged the core structural problem: they do not “have statutory authority to compel agencies to submit data in a particular format or adopt uniform definitions. As a result, much of the variation seen in the data is driven by how individual agencies interpret and implement the guidelines” [9]. Translation: each

agency has its own codebook for “use of force.” The codebooks were never aligned. The state built a channel and assumed the semantics would follow.

They did not follow. The Strogatz Inversion predicts exactly this: five years of data exchange, zero codebook convergence. The former Longmont police chief summarized it precisely: “Everyone is going to track it a little differently. That’s the problem.” The codebook is not in the signal. No amount of data exchange fixes it.

The deferred cost is not abstract. Elijah Smith was body-slammed by a Denver officer in 2023, suffered a brain bleed and nerve damage, received a settlement, and the officer was convicted and stripped of his license. The incident does not appear in the state’s use-of-force data for that year. The ambiguity was not resolved. It was paid for—in invisible harm, in broken trust, in a policy system that cannot learn from what it cannot see.

Departments that reported accurately—Aurora, Westminster, Commerce City—appear to have the worst use-of-force records in the state. The state’s own report acknowledges this: “Agencies with higher counts and rates of use of force might reflect more accurate reporting . . . as opposed to higher prevalence of force” [9]. Accurate reporters are punished. Non-reporters are invisible. The incentive structure rewards null uncertainty.

H_{total} for this system is computable. “Use of force” alone admits dozens of interpretations across agencies—different thresholds, different definitions of injury, different weapon categories. Before a single record flows, the boundary entropy is positive and measurable. No one measured it. No one built an instrument to measure it. Five years of data accumulated behind an event horizon where meaning cannot escape.

3 Definitions

Definition 1 (Codebook). *A codebook $C : \mathcal{X} \rightarrow \mathcal{R}$ is a mapping from a symbol alphabet $\mathcal{X}$ to real-world referents $\mathcal{R}$. System A operates under C_A . System B operates under C_B .*

Definition 2 (Shannon Channel). *A channel between A and B is a tuple $(\mathcal{X}, \mathcal{Y}, p(x), p(y|x))$: input alphabet, output alphabet, source distribution, and transition probability.*

Assumption (Codebook-Blind Channel). The channel $(\mathcal{X}, \mathcal{Y}, p(x), p(y|x))$ contains no variables statistically dependent on the codebook configuration. That is, the codebooks C_A and C_B do not appear in and are not correlated with any element of the tuple. If codebook-dependent variables enter the effective channel—through metadata, shared training, or side information—the channel has been expanded beyond this model, and the expansion constitutes a form of rotation.

Definition 3 (Codebook Divergence). $\delta_{AB} = d(C_A, C_B)$ for any metric d on the space of codebooks. $\delta_{AB} = 0$ iff the codebooks agree on the meaning of every symbol.

Definition 4 (Serialization Boundary). *A serialization boundary is the point at which system A writes its internal state into structured data for system B to consume. The observation function at this boundary is $O(s, \omega) = s$: the data s crosses; the codebook configuration ω does not.*

4 The Kuramoto Result (Review)

The Kuramoto model for N coupled oscillators:

$$\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i) \quad (1)$$

where θ_i is the phase of oscillator i , ω_i is its natural frequency, and K is the coupling strength.

The structural property that makes synchronization possible: the variable that must synchronize (phase θ_i) *is the signal*. The coupling term $\frac{K}{N} \sin(\theta_j - \theta_i)$ acts directly on the phase because the phase is observable through the shared medium. Strogatz (2003) showed: above critical coupling K_c , synchronization is spontaneous and passive. The medium carries the variable. Listening is sufficient.

5 The Inversion

Proposition 1 (Strogatz Inversion). *Let $(\mathcal{X}, \mathcal{Y}, p(x), p(y|x))$ be a Shannon channel between systems A and B with codebooks C_A, C_B and codebook divergence $\delta_{AB} = d(C_A, C_B)$. Then:*

- (i) *For a fixed codebook-blind channel $(\mathcal{X}, \mathcal{Y}, p(x), p(y|x))$, all channel observables are invariant to codebook divergence δ_{AB} .*
- (ii) *The channel dynamics contain no coupling force on the codebook.*
- (iii) *δ_{AB} does not converge under data exchange alone, regardless of bandwidth, duration, or volume.*

Proof. C_A and C_B do not appear in the tuple $(\mathcal{X}, \mathcal{Y}, p(x), p(y|x))$.

(i) Every channel quantity—entropy $H(X)$, mutual information $I(X; Y)$, channel capacity $C = \max_{p(x)} I(X; Y)$, error probability P_e —is a function of this tuple. The joint distribution $p(x, y)$ is identical whether $C_A = C_B$ or $C_A \neq C_B$. Therefore no channel observable distinguishes aligned from misaligned codebooks.

(ii) A coupling force requires a term in the system dynamics that is a function of the divergence δ_{AB} . Since no channel quantity is a function of δ_{AB} by (i), no such term exists. There is no force—zero, not weak.

(iii) Convergence of $\delta_{AB} \rightarrow 0$ requires a mechanism that reduces δ_{AB} over time. By (ii), no such mechanism exists in the channel dynamics. The codebook divergence at time t equals the codebook divergence at time 0 for all t , under data exchange alone. $\square$

Remark. Observed variation in practice—where downstream behavior changes with codebook mismatch—implies that codebook-dependent variables have entered the effective channel. Such expansion constitutes a form of rotation: information about the codebook has crossed the boundary, whether through metadata, shared training, or side communication. The inversion holds for the codebook-blind channel as defined.

	Kuramoto (physical)	Boundary (semantic)
Variable to sync	Phase θ_i	Codebook C_i
In the signal?	Yes	No
Coupling force	$\frac{K}{N} \sin(\theta_j - \theta_i)$	None
Convergence	Spontaneous above K_c	Does not occur
Recovery	Passive (listen)	Active (rotation)

Every row inverts. This is not an analogy. It is a structural reversal. Strogatz proved that when the variable is in the signal, synchronization is free. The inversion proves that when the variable is not in the signal, synchronization is impossible without active measurement.

6 The Closure Property

The inversion rests on a more elementary fact.

Lemma 1 (Closure). *Let S be a system’s state space. Let $F(S)$ denote the class of functions computable over S . For any variable $\omega \notin S$: no $f \in F(S)$ outputs the value of ω .*

Proof. f takes inputs from S . $\omega \notin S$. Therefore ω is not in the domain of f . $\square$

This is close to a tautology. That is the point—a two-line proof that cannot be argued with is stronger than a deep proof with assumptions to attack. We address why near-tautology is the feature, not the limitation, in the companion paper [6].

Corollary 1 (Diagnostic Blindness). *Every diagnostic, residual analysis, specification test, and anomaly detector is a function of S . When all report success, they confirm consistency within S . They cannot confirm that S is complete.*

Proof. Each is an $f \in F(S)$. Apply Lemma 1. $\square$

7 The Boundary Entropy Bound

Definition 5 (Boundary Entropy). *Let $F_{\text{ambiguous}}$ be the set of fields crossing a serialization boundary for which $|\Omega_i| > 1$ interpretations exist. The boundary entropy is:*

$$H_{\text{total}} = \sum_{i \in F_{\text{ambiguous}}} \log_2 |\Omega_i| \quad (2)$$

This is an upper bound under maximum ignorance (uniform distribution over interpretations). Non-uniform priors—derived from past rotations or domain knowledge—may reduce effective uncertainty below this bound. H_{total} decomposes per field; partial alignment reduces the sum without eliminating it.

Proposition 2 (Conservation). *H_{total} is a lower bound on boundary uncertainty. No computation on the received data reduces it. The only operation that reduces H_{total} is the acquisition of bits from across the boundary.*

Proof. The ambiguity arises because the codebook did not cross the boundary (Definition 4). The received data s is identical across all $\omega \in \Omega$ (the observation function $O(s, \omega) = s$ projects away ω). Every computation on s returns the same output for every ω . Therefore no computation on s distinguishes configurations, and H_{total} is invariant under $F(S)$.

Bits from across the boundary—acquired through rotation—reduce $|\Omega_i|$ by eliminating configurations. Any reduction in codebook ambiguity requires, in its causal history, at least one such cross-boundary acquisition. No computation on data already present in the system’s state space substitutes for it. $\square$

8 Rotation as the Irreducible Unit

Definition 6 (Rotation). *A rotation is the act of an observer at a system boundary shifting orientation from inward (own codebook) to outward (cross-boundary signal), acquiring information, recording the return, and comparing.*

Rotation may occur at any point in the causal history of a system. A system at a boundary may rely on the results of rotations performed in the past—during training, during integration, during a conversation between two engineers ten years ago. The claim is not that rotation must happen at the moment of crossing. The claim is that *no reduction in codebook ambiguity can occur without some rotation occurring somewhere in the system’s causal history*. Every source of alignment traces back, eventually, to an observation made across a boundary.

We call the stateful node that performs this operation the *Observatron* (Figure 1). At each timestep, it occupies one of two orientations—inward (consulting its own codebook) or outward (acquiring the cross-boundary signal)—and encounters exactly one of three decisions: **Act** (codebooks aligned, proceed), **Ask** (mismatch detected, acquire information), or **Halt** (boundary outside reasoning capacity). Neither side sees both frames simultaneously. The rotation between frames is the irreducible unit of measurement.

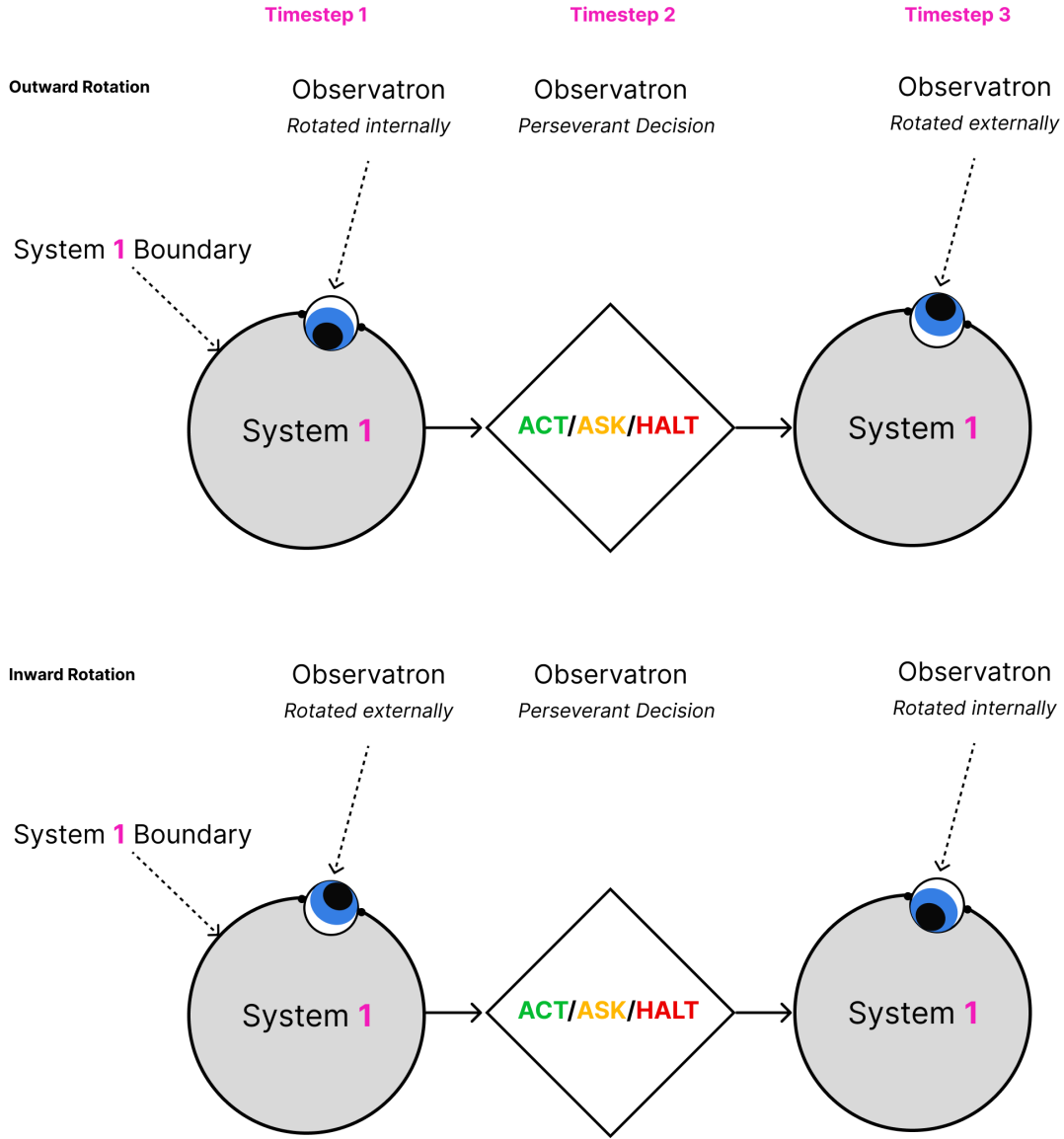


Figure 1: The Observatron: a stateful boundary node that rotates between inward (own codebook) and outward (cross-boundary signal) orientations. At each timestep, the observer shifts orientation, encounters the Act/Ask/Halt decision, and records the return. Neither side sees both frames simultaneously. Top: outward rotation. Bottom: inward rotation.

Any mechanism that resolves boundary ambiguity must instantiate, somewhere in its causal history, a cross-boundary acquisition of codebook information. We call this operation rotation. What is new is not the observation that boundaries require measurement—skilled engineers have performed informal rotations for decades. What is new is the decision interface:

every boundary must evaluate into exactly one of **Act** (codebooks aligned, proceed), **Ask** (mismatch detected, acquire information), or **Halt** (boundary outside reasoning capacity). This interface is exhaustive, mutually exclusive, protocol-enforceable, and composable across networked systems (Proposition 2). Rotation is:

- **Necessary:** any mechanism that achieves codebook alignment must include, in its causal history, at least one cross-boundary acquisition of codebook information. Learning, conventions, and shared priors are accumulations of prior rotations, not alternatives to them.
- **Costly:** each rotation requires at minimum the acquisition and comparison of H_{total} bits. By Landauer’s principle, each bit requires at minimum $kT \ln 2$ joules. The thermodynamic floor is real but negligible; the practical floor—compute, attention, organizational time—is orders of magnitude higher.
- **Non-eliminable:** no architecture, no protocol, no amount of data exchange removes the need for rotation. It can only redistribute where and when the cost is paid.

9 Why Learning Does Not Eliminate Rotation

A system that has been trained on labeled data has consumed the results of rotations performed by its labelers. A system that relies on shared conventions consumes the results of rotations performed by the communities that established those conventions. A system that infers codebook alignment from statistical regularities in past exchanges is consuming the results of implicit rotations—observations that happened to cross the boundary and happened to be informative.

None of these are alternatives to rotation. They are *accumulations of prior rotations*—compressed, stored, and reused.

Apparent alternative	Actual source
Labeled training data	Rotations by annotators
Shared conventions	Historical rotations by communities
Statistical convergence	Implicit rotations via informative exchange
Shared priors	Compressed prior rotations
ML models	Trained on products of rotation

The question is never *whether* rotation occurs. It is *when* it occurred, *who* paid for it, and *whether it is still valid*—because codebooks drift. The learning loop is developed fully in [7], Section 8.4; the formal definition of rotation and its cost structure appear in [7], Section 2.5.

10 The Deferred Cost of Ambiguity

Proposition 3 (Deferred Cost). *Let $H_{\text{total}} > 0$ at a serialization boundary. If no cross-boundary information is acquired (no rotation, past or present), then the system must incur a non-zero expected loss due to misinterpretation. This loss may be delayed, distributed, or hidden, but it cannot be eliminated.*

Proof. By Proposition 2, H_{total} is invariant under $F(S)$. Let $P(\omega)$ be any prior over Ω . The system selects one configuration ω^* and acts on it. Since $H_{\text{total}} > 0$, $|\Omega| > 1$, so there exists at least one $\omega \neq \omega_{\text{true}}$ with $\Delta(\omega, \omega_{\text{true}}) = |V^*(\omega) - V^*(\omega_{\text{true}})| > 0$ (by the assumption that different codebook configurations produce different optimal actions). The expected loss is:

$$\mathbb{E}[\mathcal{L}] = \sum_{\omega \neq \omega_{\text{true}}} P(\omega) \cdot \Delta(\omega, \omega_{\text{true}}) \quad (3)$$

Unless $P(\omega_{\text{true}}) = 1$ —which requires that the system has already collapsed H_{total} to zero through prior rotation— $\mathbb{E}[\mathcal{L}] > 0$. Priors derived from past rotations may reduce expected loss, but cannot eliminate it without collapsing H_{total} to zero. If $H_{\text{total}} > 0$, some residual loss remains. $\square$

Ambiguity does not disappear. It is either resolved or paid for. Boundary entropy behaves as a conserved cost: it can be reduced by acquiring information or expressed as expected loss, but not eliminated without payment.

A system may tolerate ambiguity—many do, successfully—by accepting residual error within operational tolerances. The claim is not that unresolved ambiguity always causes catastrophic failure. The claim is that perfect alignment requires paying the entropy cost, and that the entropy cost is computable before any agent runs.

The three options at any boundary:

1. **Pay now** (Ask). Perform rotation. Acquire information. Reduce H_{total} . Pay in time, compute, and attention.
2. **Pay later** (Act without rotation). Proceed on assumption. Accept expected loss $\mathbb{E}[\mathcal{L}] > 0$. Pay in errors, rework, and operational failure.
3. **Refuse to pay** (Halt). Decline to act. Pay in missed opportunity.

There is no fourth option. There is no free lunch on semantic uncertainty.

A rational system minimizes total boundary cost:

$$\mathcal{C}_{\text{total}} = \mathcal{C}_{\text{rotation}} + \mathbb{E}[\mathcal{L}_{\text{residual}}] + \mathcal{C}_{\text{opportunity}} \quad (4)$$

where $\mathcal{C}_{\text{rotation}}$ is the cost of cross-boundary measurement, $\mathbb{E}[\mathcal{L}_{\text{residual}}]$ is the expected loss from unresolved ambiguity, and $\mathcal{C}_{\text{opportunity}}$ is the cost of inaction. Each decision redistributes cost across terms:

Decision	$\mathcal{C}_{\text{rotation}}$	$\mathbb{E}[\mathcal{L}_{\text{residual}}]$	$\mathcal{C}_{\text{opportunity}}$
Ask	Increases	Decreases	0
Act	0	Increases	0
Halt	0	0	Increases

No decision zeroes all three terms simultaneously when $H_{\text{total}} > 0$. The total cost is bounded below by the boundary entropy: some payment is unavoidable. The only choice is the currency.

11 The Defensive Dual

Corollary 2 (Semantic Firewall). *If codebook alignment cannot propagate passively through the channel, then codebook contamination cannot propagate passively through the channel either.*

Proof. Contamination of C_B by C_A requires a mechanism by which C_A modifies C_B through the channel. By Proposition 1(ii), no channel quantity couples to the codebook. Therefore no channel-mediated mechanism modifies C_B . The serialization boundary is a semantic firewall: opaque in both directions. $\square$

12 Falsification Criteria

Each claim is paired with the condition under which it would be withdrawn.

1. **Proposition 1 (Strogatz Inversion):** Produce a channel observable that varies with δ_{AB} , or demonstrate codebook convergence under data exchange alone without any cross-boundary query.
2. **Lemma 1 (Closure):** Produce a computation on locally available data that identifies and resolves a specific missing dimension without acquiring information from outside the state space.
3. **Proposition 2 (Conservation):** Produce a computation on the received data alone that reduces $|\Omega_i|$ for any ambiguous field without information—past or present—from the sender.
4. **Proposition 3 (Deferred Cost):** Produce a system that achieves zero expected loss at a boundary with $H_{\text{total}} > 0$ without any rotation in its causal history.
5. **Rotation necessity:** Produce a source of codebook alignment information that does not trace back to a cross-boundary observation.

13 Scope and Limitations

The closure property is close to a tautology. Its strength and limitation are the same: the proof is trivial; the consequence—that every deployed diagnostic is blind to codebook misalignment—is not.

The Strogatz Inversion is *structural*, not analogical. It does not claim that decision systems are oscillators. It claims that the structural relationship between the synchronization variable and the signal *inverts* between physical coupling (variable in signal) and semantic coupling (variable not in signal), and that this inversion has a provable consequence: zero convergence.

The boundary entropy H_{total} depends on the enumeration of $|\Omega_i|$, which is a modeling choice. The claim is not that any specific value of H_{total} is derived from first principles. The claim is that $H_{\text{total}} > 0$ for any boundary where at least one field admits more than one interpretation—and that this bound is computable before any agent runs.

The Deferred Cost proposition holds for any prior over Ω , not only the maximum-ignorance case. Expected loss is strictly positive unless $P(\omega_{\text{true}}) = 1$, which requires that H_{total} has already been collapsed to zero through prior rotation. Systems with informative priors—derived from past rotations—may have lower expected loss, but cannot eliminate it while $H_{\text{total}} > 0$.

Semantic standards—RDF, OWL, SHACL, ontology alignment—are powerful and valuable. They work for centralized control of system-to-system mappings where top-down authority and specialist knowledge engineers are available. They do not scale to liquid boundaries where variables change continuously, no central authority governs the crossing, and agent-paced operation demands thousands of verifications per hour. What is missing is not vocabulary but a *protocol*: a lightweight runtime mechanism where systems verify their own codebook alignment at the moment of crossing. The Observatron and the rotation it performs are the minimal instrument for that verification.

14 Toward a Resolution Architecture

This paper establishes the problem. The companion paper—*Liquid Hypergraphs: A Context Graph Protocol* (Itelman & Kowalski, 2026)—proposes the architecture to resolve it.

The core claim of this paper, stated as a single sentence: *if the codebook is not in the signal, then its uncertainty cannot be reduced without cross-boundary information, and unresolved uncertainty manifests as cost*. The Strogatz Inversion proves the first clause. The Conservation proposition proves the second. The Deferred Cost proposition proves the third.

What follows from this claim is an architectural requirement. If every serialization boundary carries dark uncertainty, and if that uncertainty can only be reduced by active measurement at the boundary, then the architecture that resolves it must place observers *at* the boundaries—not in the systems on either side. We call these observers Observatrons: stateful boundary nodes that rotate between inward orientation (own codebook) and outward orientation (cross-boundary signal), encounter a decision (Act, Ask, or Halt), and record the result.

The analogy to coupled oscillators is not merely illustrative—it is architectural. In Kuramoto’s model, synchronization emerges because every oscillator is simultaneously a sensor and a signal source: it listens to the shared medium and contributes to it. The Strogatz Inversion shows that semantic synchronization cannot emerge passively. But the architectural pattern still applies: what is needed is a flat mesh of boundary observers, each with its own decision interface, each capable of detecting desynchronization and self-correcting through a shared protocol—the way a musician uses a metronome and a shared scale to sense when they are out of sync and adjust.

Context Graphs **REQUIRE**
a "liquid" architecture...

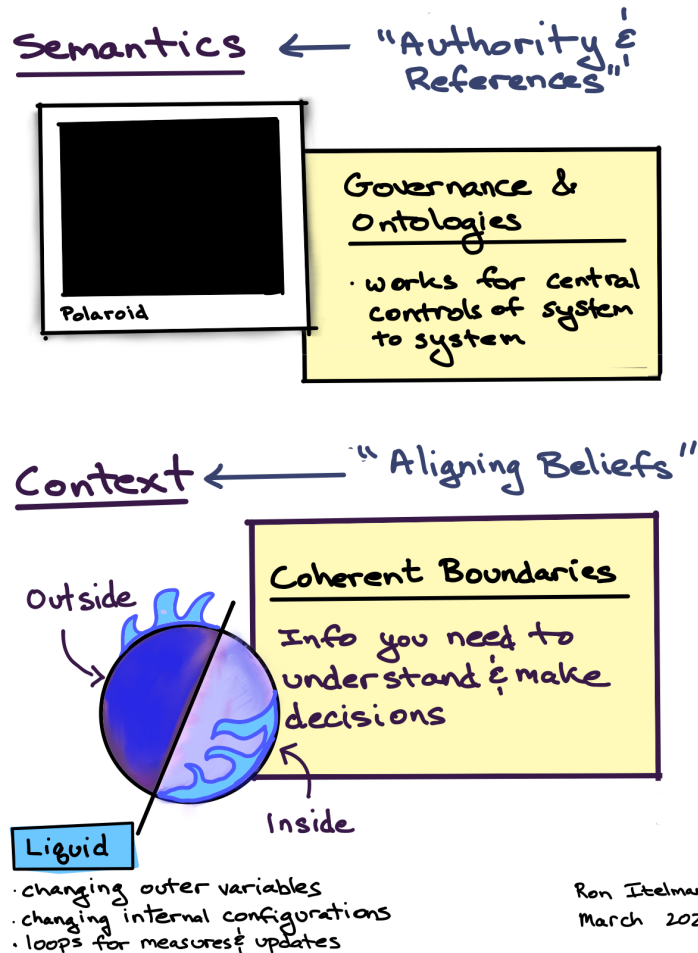


Figure 2: Semantics vs. Context as architectural modes. Semantics (top) operates through authority and references—governance, ontologies, centralized control of system-to-system mappings. Context (bottom) operates through aligning beliefs at coherent boundaries—liquid, continuously changing outer variables and internal configurations, requiring loops for measurement and updates. Context graphs require a liquid architecture.

The Liquid Hypergraphs paper formalizes this architecture: a dependency-ordered four-facet gauge (Context, Meaning, Structure, Data) producing three reasoning modes (Halt, Ask, Act), recorded as canonical claims in a five-column form with dereferenceable URIs.

The liquid hypergraph is the data structure that accumulates the results: its edges exist because measurements were made, and its absent edges represent unmeasured boundaries—visible gaps, not hidden ones. The coherence coverage metric $C = |E|/N$ makes the organization's unmeasured exposure structurally visible. The protocol implements the measurement. The graph records the history. The metric tracks the progress. Together, they provide the

always-on sensor network that navigates system boundaries—not the ships that sail between them, but the buoys on the ocean.

The protocol begins in what we call the *trivial regime*: Boolean comparisons over minimal structure—four facets, five columns, URI matching. This regime is deliberately simple, designed to exhaust deterministic alignment before introducing probabilistic machinery. Beyond it lies the *probabilistic regime*, where Fisher information governs the cost of resolving continuous codebook spaces—connecting to Kowalski’s independent work on hierarchical metric flow [7]. It is our hypothesis that the Act/Ask/Halt decision interface can operate not only between systems but *within* them—embedded inside neural network architectures as boundary-aware layers. This is future work.

The canonical claim form is designed to support real-time, on-edge translation into tensor representations compatible with Domingos’ tensor logic [?]; we formalize this bridge in JSON-TL [?]. Each Observatron carries a unique dereferenceable identity (the `source` column of the canonical claim form), enabling flat projections onto a shared gauge while preserving node-level provenance—a requirement for well-defined network dynamics [1].

15 Conclusion

A decision modeler can get their model right. The Bellman equation can be perfectly specified within a single system. But no decision system operates alone, and at the boundaries of the network, the codebook does not cross with the data. The downstream system’s state space desynchronizes from reality. The value function converges—to a number that may not correspond to anything in the world. Every diagnostic reports success.

This is dark uncertainty. It is structurally present at every serialization boundary. It is computationally consequential—in our demonstration, it produces a $4\times$ variance in execution from identical input. And it is invisible to every instrument that operates within the system, because the variable is not in the state space.

Strogatz proved that coupled oscillators synchronize for free because the phase is in the signal. We prove the structural inverse: at serialization boundaries, the codebook is not in the signal. It was never in the signal. There is no standard that puts it there. Alignment requires active, costly, irreducible measurement—rotation—at the boundary.

Learning does not eliminate rotation. It consumes the products of prior rotations. Shared conventions do not eliminate rotation. They are the residue of historical rotations. The question is never whether rotation occurs but when, who paid, and whether it is still valid.

Unresolved ambiguity does not disappear. It is either resolved or paid for—in errors, in rework, in operational failure, in trades executed on a Sunday when the exchange is closed, in five years of police reporting data that means nothing because no one aligned the codebooks. The cost has a computable floor. No architecture eliminates it. You can only choose where you pay.

Dark uncertainty deserves to be formalized. This paper is a first step.

All claims carry falsification criteria. We invite correction.

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